United States Patent Application Publication	20250268740
Kind Code	A1
Publication Date	August 28, 2025
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CERVICAL IMMOBILIZATION DEVICE

Abstract

A cervical immobilization device (CID) includes a front collar. The front collar includes a chest brace configured to extend downward from both shoulders of a patient and rest against a chest of the patient. An adjustable headband is configured to strap around a forehead of the patient. First and second lateral head supports are configured to attach and extend vertically downward from the headband. The first and second lateral head supports are configured to form a rigid connection to the front collar. First and second underarm straps each have first and second ends. The first ends are configured to attach to opposing sides of the chest brace below the patient's shoulders. The second ends are configured to attach to opposing sides of the CID above the patient's shoulders when the CID is worn by the patient.

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Appl. No.: **19/062679**

Filed: **February 25, 2025**

Related U.S. Application Data

us-provisional-application US 63557801 20240226

Publication Classification

Int. Cl.: **A61F5/055** (20060101)

U.S. Cl.:

CPC **A61F5/055** (20130101);

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a non-provisional application of, and claims the benefit of the filing date of, U.S. provisional application 63/557,801, filed Feb. 26, 2024, entitled "CERVICAL IMMOBILIZATION DEVICE," the contents of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to cervical immobilization devices. More specifically, the disclosure relates to cervical immobilization devices that include a front collar with a chest brace, wherein a pair of underarm straps attach to the chest brace below the patient's shoulders and attach to the front collar above the patient's shoulders.

BACKGROUND

[0003] Cervical immobilization devices (CIDs), or cervical neck braces, are used to substantially immobilize an injured patient in the field. However, prior art CIDs often have multiple components that are difficult to attach to a patient, especially if the patient is unconscious. Furthermore, prior art CIDs do not adequately immobilize the patient even when properly applied. Moreover, such prior art CIDs are often uncomfortable for the patient to wear.

[0004] Additionally, prior art CIDs often have chin bracing components which are positioned around the chin area to further immobilize the head. Problematically however, such chin bracing components make it difficult for a first responder to intubate a patient when needed, because they are not easily removable and substantially impair chin movement. As a result, prior art CIDs that brace the chin often have to be completely removed when a patient is being intubated, which unnecessarily heightens the risk of further injury to the patient.

[0005] Accordingly, there is a need for a cervical immobilization device which is more comfortable to wear, is more easily attached and adjusted in the field, and better immobilizes the head of a patient than prior art CIDs. Additionally, there is a need for a cervical immobilization device which enables a patient to be intubated without having to remove substantially all of the CID.

BRIEF DESCRIPTION OF THE INVENTION

[0006] The present disclosure offers advantages and alternatives over the prior art by providing a cervical immobilization device (CID), which is easily attachable and adjustable in the field and is comfortable to wear. The CID includes a front collar with a chest brace that is rigidly connected to a headband to immobilize forward movement of the head.

[0007] Additionally, adjustable first and second underarm straps are connected at their distal ends directly to the chest brace below the patient's shoulder and directly to the front collar above the patient's shoulder. The first and second underarm straps are configured to extend under and abut against the floor of the armpit so that, when tightened snug against the patient's armpits, the underarm straps prevent the head of the patient from moving rearward.

[0008] Moreover, the CID of the present disclosure may include a chin brace that is removably attached to the front collar. The chin brace is configured to be the sole component of the CID required to be removed from the CID to enable a patient to be intubated while wearing the CID.

[0009] A cervical immobilization device in accordance with one or more aspects of the present disclosure includes a front collar. The front collar includes a chest brace configured to extend downward from both shoulders of a patient and rest against a chest of the patient. An adjustable headband is configured to strap around a forehead of the patient. First and second lateral head supports are configured to attach and extend vertically downward from the headband. The first and second lateral head supports are configured to form a rigid connection to the front collar. First and second underarm straps each have first and second ends. The first ends are configured to attach to opposing sides of the chest brace below the patient's shoulders. The second ends are configured to attach to opposing sides of the CID above the patient's shoulders when the CID is worn by the patient.

[0010] Another cervical immobilization device in accordance with one or more aspects of the present disclosure includes a front collar. The front collar includes a chest brace configured to extend downward from both shoulders of a patient and rest against a chest of the patient. An adjustable headband is configured to strap around a forehead of the patient. First and second lateral head supports are configured to attach and extend vertically downward from the headband. The first and second lateral head supports are configured to form a rigid connection to the front collar. A chin brace is removably attached to opposing sides of the front collar. The chin brace is configured to provide additional support to immobilize the patient, when the CID is being worn by the patient. The chin brace is configured to be the sole component of the CID required to be removed from the CID to enable the patient to be intubated while wearing the CID.

[0011] Another cervical immobilization device in accordance with one or more aspects of the present disclosure includes a front collar. The front collar includes a single integral structure, which includes the structural features of a chest brace, first and second shoulder pieces and first and second collar projections. The chest brace is configured to extend downward from both shoulders of a patient and rest against a chest of the patient. The first and second shoulder pieces are integrally attached to opposing upper ends of the chest brace. The first and second shoulder pieces are configured to project over and rest upon the shoulders of the patient to support the front collar thereon. The first and second lateral collar projections are integrally attached to opposing upper ends of the chest brace. The first and second lateral collar projections extend vertically upwards therefrom. An adjustable headband is configured to strap around a forehead of the patient. First and second lateral head supports are configured to attach and extend vertically downward from the headband. The first and second lateral head supports are configured to adjustably and rigidly attach to the first and second lateral collar projections to adjust a distance between the headband and the front collar. First and second underarm straps each having first and second ends. The first end of the first underarm strap is configured to attach to a left side of the chest brace below the patient's shoulder and the second end of the first underarm strap is configured to attach to a right side of the CID above the patient's shoulder. The first end of the second underarm strap is configured to attach to a right side of the chest brace below the patient's shoulder and the second end of the second underarm strap is configured to attach to a left side of the CID above the patient's shoulder.

[0012] It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail below (provided such concepts are not mutually inconsistent) are contemplated as being part of the inventive subject matter disclosed herein and may be used to achieve the benefits and advantages described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The disclosure will be more fully understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0014] FIG. **1** depicts an example of a front perspective view of a cervical immobilization device, according to aspects described herein;

[0015] FIG. **2** depicts an example of a rear perspective view of the cervical immobilization device of FIG. **1** being worn by a patient, according to aspects described herein;

[0016] FIG. **3** depicts an example of a perspective view of the cervical immobilization device of FIG. **1** being worn by a patient, according to aspects described herein;

[0017] FIG. **4** depicts an example of a perspective view of a front collar with a chest brace of the cervical immobilization device of FIG. **1**, according to aspects described herein;
[0018] FIG. **5** depicts an example of a perspective view of a headband and first and second lateral head supports of the cervical immobilization device of FIG. **1**, according to aspects described herein;
[0019] FIG. **6** depicts an example of a chin brace of the cervical immobilization device of FIG. **1**, according to aspects described herein;
[0020] FIG. **7**A depicts an example of a partially exploded view of the cervical immobilization device of FIG. **1** being worn by a patient with the chin brace separated from the cervical immobilization device, according to aspects described herein; and
[0021] FIG. **7**B depicts an example of a perspective view of the cervical immobilization device of FIG. **1** being worn by a patient without a chin brace while being intubated, according to aspects described herein.

DETAILED DESCRIPTION OF THE INVENTION

[0022] Certain examples will now be described to provide an overall understanding of the principles of the structure, function, manufacture, and use of the methods, systems, and devices disclosed herein. One or more examples are illustrated in the accompanying drawings. Those skilled in the art will understand that the methods, systems, and devices specifically described herein and illustrated in the accompanying drawings are non-limiting examples and that the scope of the present disclosure is defined solely by the claims. The features illustrated or described in connection with one example may be combined with the features of other examples. Such modifications and variations are intended to be included within the scope of the present disclosure.
[0023] The terms “significantly”, “substantially”, “approximately”, “about”, “relatively,” or other such similar terms that may be used throughout this disclosure, including the claims, are used to describe and account for small fluctuations, such as due to variations in processing from a reference or parameter. Such small fluctuations include a zero fluctuation from the reference or parameter as well. For example, they can refer to less than or equal to ±10%, such as less than or equal to ±5%, such as less than or equal to ±2%, such as less than or equal to ±1%, such as less than or equal to ±0.5%, such as less than or equal to ±0.2%, such as less than or equal to ±0.1%, such as less than or equal to ±0.05%.
[0024] Referring to FIGS. **1** and **2**, an example is depicted of a front perspective view off of a patient **122** (FIG. **1**) and a rear perspective view when being worn by a patient **122** (FIG. **2**) of a cervical immobilization device (CID) **100**, according to aspects described herein. The CID includes a front collar **102**, an adjustable headband **104**, a first lateral head support **106** and a second lateral head support **108**. The CID may also include a chin brace **110**. The front collar **102** includes a chest brace **112**, a first shoulder piece **114**, a second shoulder piece **116**, a first lateral collar projection **118** and a second lateral collar projection **120**.
[0025] As mentioned above, the front collar **102** of the CID **100** includes the chest brace **112**. The chest brace is configured to extend downward from both shoulders of a patient **122** (see FIG. **3**) and rest against a chest **124** of the patient **122**. The chest brace **112** may be made of a hard plastic material of any other appropriately rigid material. The chest brace **112** may also have padding **128** disposed thereon for the comfort of the patient.
[0026] The adjustable headband **104** of the CID **100** is configured to strap around a forehead **126** of the patient **122**. The headband is illustrated herein as utilizing hook and loop fasteners to adjustably fasten to the forehead **126** of the patient **122**. However, any number of appropriate fastening systems may be used. For example, buckles, snaps, tabs or other fastening devices may be used to enable adjustment to the headband around a patient's head.
[0027] The first and second lateral head supports **106**, **108** are configured to attach and extend vertically downward from the headband **104**. The lateral head supports may be made of a hard plastic or any appropriate rigid material. As illustrated, the lateral head supports **106**, **108** each have

a head section **130**, with two parallel headband slots **132** for the headband **104** to feed through and attach thereto. However, the lateral head supports **106**, **108** may be attached to the headband **104** via any number of appropriate attachment systems. For example, buckles, snaps, tabs or other fastening devices may be used. Additionally, the lateral head supports **106**, **108** may be integrally or permanently attached to the headband **104**.

[0028] The first and second lateral head supports **106**, **108** are configured to form a rigid connection to the front collar **102**. As will be discussed in greater detail herein, the rigid connection illustrated in FIGS. **1** and **2** is the direct connection between the first and second lateral head supports **106**, **108** and the associated first and second lateral collar projections **118**, **120** of the front collar **102**. However, the lateral head supports **106**, **108** may be rigidly connected to the collar **102** by any number of appropriate connections either directly or indirectly. For example, the lateral head supports **106**, **108** may be rigidly attached to a chin brace, similar to the chin brace **110**. The chin brace may then be rigidly connected to the chest brace **112** of the collar **102** to provide an indirect rigid connection between the lateral head supports **106**, **108** and the collar **102**.

[0029] The CID **100** also includes a first underarm strap **134** and a second underarm strap **136**. The first underarm strap **134** has a first end **138** and a second end **140**. The second underarm strap **136** also has a first end **142** and a second end **144**. The first ends **138**, **142** of each underarm strap **134**, **136** are configured to attach to a left side **146** and an opposing right side **148** of the chest brace **112** below the patient's shoulders **150**. The first ends **138**, **142** may also connect to a lower end portion **152** of the chest brace **112**.

[0030] The underarm straps **134**, **136** may be removably attached to the CID via any number of appropriate connection devices. For example, the underarm straps **134**, **136** may be attachable at either end using quick slide and release buckles as illustrated in FIGS. **1** and **2**. Alternatively, the underarm straps **134**, **136** may be removable and/or attachable utilizing a hook and loop fastener system. Moreover, the underarm straps **134**, **136** may be removably attachable at only one end and permanently attached at the other end (as illustrated in FIGS. **1** and **2**), or the underarm straps **134**, **136** may be removably attachable at both ends.

[0031] The second ends **140**, **144** of each underarm strap **134**, **136** are configured to attach to opposing sides of the CID **100** above the patient's shoulders **150** when the CID **100** is worn by the patient **122** (see FIG. **3**). For example, the second ends **140**, **144**, may be attached a left side and a right side of the collar **102** of the CID **100** above the shoulders **150** of the patient **122**. More specifically, the second ends **140**, **144** may be attached to an left upper end **154** and an opposing right upper end **156** of the chest brace **112** of the front collar **102**.

[0032] Though the underarm straps **134**, **136** are illustrated herein as being attached to the front collar **102** of the CID **100** above the patient's shoulders **150**, it is within the scope of the present disclosure that the underarm straps **134**, **136** may also attach to other components of the CID **100**. For example, in an alternative embodiment, the lateral head supports **106**, **108** of the CID **100** may be configured to attach to the second ends **140**, **142** of the underarm straps **134**, **136**.

[0033] The first and second underarm straps **134**, **136** may be configured to extend under and abut against a floor of the patient's armpits **158** when the CID **100** is worn by the patient **122** (see FIG. **3**). Armpit, as used herein, means the hollow beneath the junction of the arm and shoulder of a patient **122**. More specifically, the armpit includes the pyramid shaped part of the body between the upper arm and the side of the thorax of a patient **122**. The armpit is floored by the skin of the armpit.

[0034] The first and second underarm straps **134**, **136**, may extend from the front of the patient **122** to the rear of the patient without crisscrossing in the back of the patient **122**. However, for added stability, the underarm straps **134**, **136** may be made to crisscross at the back of the patient **122**. More specifically, the first end **138** of the first underarm strap **134** may be configured to attach to the left side **146** of the chest brace **112** below the patient's shoulder **150** and the second end **140** of the first underarm strap **134** may be configured to attach to the right side of the CID **100** above the patient's shoulder **150**. Additionally, the first end **142** of the second underarm strap **136** may be configured to attach to a right side **148** of the chest brace **112** below the patient's shoulder **150** and the second end **144** of the second underarm strap **136** may be configured to attach to a left side of the CID **100** above the patient's shoulder **150**.

[0035] The front collar **102** may also include the first and second shoulder pieces **114**, **116**. The first and second shoulder pieces **114**, **116** are integrally attached to the left upper end **154** of the chest brace **112** and the opposing right upper end **156** of the chest brace **112** of the front collar **102**. The first and second shoulder pieces **114**, **116** are configured to project over and rest upon the shoulders **150** of the patient **122** to support the front collar **102** thereon. The first and second shoulder pieces **114**, **116** provide a positive stop for the front collar **102**, while the front collar **102** advantageously provides support for the positioning of the other components of the CID **100** (e.g., the first and second lateral supports **106**, **108**, the headband **104** and the chin brace **110**).

[0036] The front collar **102** may also include the first and second lateral collar projections **118**, **120**. The first and second lateral collar projections **118**, **120** are also integrally attached to opposing upper ends **154**, **156** of the chest brace **112**. The first and second lateral collar projections **118**, **120** extend vertically upwards from their associated upper ends **154**, **156** of the chest brace **112**. The first and second lateral collar projections **118**, **120** are configured to provide a rigid but adjustable attachment to the first and second lateral head supports **106**, **108**. The adjustable attachment of the lateral collar projections **118**, **120** to the lateral head supports **106**, **108** enables a distance **160** between the headband **104** and the front collar **102** to be adjusted. More specifically, the lateral collar projections **118**, **120** may adjust a distance **160** between the headband **104** and the first and second shoulder pieces **116**, **118** of the front collar **102**. The adjustability of the distance **160** enables the CID **100** to accommodate various neck lengths and head sizes of the patient **122**.

[0037] The lateral collar projections **118**, **120** may be adjustably attached to the lateral head supports **106**, **108** by any number of appropriate attachment systems. For example, by a hook and loop attachment system, or by utilizing a system of buckles.

[0038] However, the attachment system illustrated in FIGS. **1** and **2** utilizes a system of adjustment tabs **162** positioned on the lateral head supports **106**, **108** and adjustment thru-holes **164** positioned on the lateral collar projections **118**, **120**. More specifically, a plurality of headband adjustment tabs **162** may be positioned vertically on the first and second lateral head supports **106**, **108**. Additionally, a plurality of headband adjustment thru-holes **164** may be positioned vertically on the first and second lateral collar projections **118**, **120**. The headband adjustment thru-holes **164** are sized to receive the headband adjustment tabs **162**. During operation, one or more of the plurality of headband adjustment tabs **162** mate with (or press fit into) one or more of the plurality of headband adjustment thru-holes **164** to provide a rigid, but adjustable, connection between the first and second lateral head supports **106**, **108** and the first and second lateral collar projections **118**, **120** respectively. The adjustable connection enables the adjustment of the distance **160** between the headband **104** and the shoulder projections **114**, **116** of the front collar **102**.

[0039] Alternatively, the attachment system may also utilize a system of adjustment tabs **162** positioned on the lateral collar projections **118**, **120** and adjustment thru-holes **164** positioned on the lateral head supports **106**, **108**. More specifically, a plurality of headband adjustment tabs **162** may be positioned vertically on the first and second lateral collar projections **118**, **120**. Additionally, a plurality of headband adjustment thru-holes **164** may be positioned vertically on the first and second head supports **106**, **108**. The operation of this embodiment of an attachment system with the adjustment tabs **162** positioned on the lateral collar projections **118**, **120** remains essentially the same as the embodiment of the attachment system with the adjustable tabs **162** positioned on the lateral head supports **106**, **108**.

[0040] The CID may also include the chin brace **110**. The chin brace **110** is removably attached to opposing sides of the front collar **102**. The chin brace **110** is configured to provide additional support to immobilize the patient **122**, when the CID **100** is being worn by the patient **102**.

Advantageously, the chin brace **110** is configured to be the sole component of the CID **100** that is required to be removed from the CID **100** to enable the patient **122** to be intubated or perform other medically necessary interventions while still wearing the CID **100**.

[0041] As illustrated in FIGS. **1** and **2**, the chin brace **110** is pivotally and adjustably attached to opposing sides of the front collar **102** to enable a pivot angle **166** between the chin brace **110** and the front collar **102** to be adjusted. More specifically, the pivot angle **166** may be between the chin brace **110** and the chest brace **112** of the front collar **102**.

[0042] The chin brace **110** may be made removably attachable and/or pivotably attachable to the front collar **102** through any number of appropriate attachment systems. However, in the chin brace **110** illustrated in FIGS. **1** and **2**, the attachment system of the chin brace **110** includes a first forked chin brace end portion **168** and a second forked chin brace end portion **170**. Each forked chin brace end portion **168**, **170** includes a first pivoting chin brace extension **172** pivotally and removably attached to the front collar **102**. Additionally, each forked chin brace end portion **168**, **170** includes a second adjusting chin brace extension **174** adjustably and removably attached to the front collar **102**.

[0043] More specifically, each first pivoting chin brace extension **172** includes one of a first tab **176** or a first thru-hole **178**, which is configured to mate with the other of the first tab **176** or the first thru-hole **178** positioned on the front collar **102**. In other words, if the first tab **176** is positioned on the first pivoting chin brace extension **172**, then the first thru-hole **178** will be positioned on the front collar **102**. Alternatively, if the first tab **176** is positioned on the front collar **102**, then the first thru-hole **178** will be positioned on the first pivoting chin brace extension **172.** In either case, the first tab **176** is configured to pivot within the first thru-hole **178** to enable the chin brace **110** to pivot relative to the front collar **102**.

[0044] Additionally, each second adjusting chin brace extension **174** includes one of a plurality second tabs **180** or a plurality of second thru-holes **182**, which are configured to mate with the other of the plurality of second tabs **180** or the plurality of second thru-holes **182** positioned on the front collar **102**. In other words, if the plurality of second tabs **180** are positioned on the second adjusting chin brace extension **174**, then the plurality of second thru-holes **182** will be positioned on the front collar **102**. Alternatively, if the plurality of second tabs **180** are positioned on the front collar **102**, then the plurality of second thru-holes **182** will be positioned on the second adjusting chin brace extension **174**. In either case, one or more of the plurality of second tabs **180** mate with one or more of the plurality of second thru-holes **182** to provide an adjustable connection between the second adjusting chin brace extensions **174** and the front collar **102** to enable adjustment of the pivot angle **166** between the chin brace **110** and the front collar **102**.

[0045] Referring to FIG. **3**, an example is depicted of a perspective view of the cervical immobilization device **100** being worn by the patient **122**, according to aspects described herein. During operation, the CID **100** may be easily fit to the patient **122** by placing the front collar **102** over the patient's chest **124** and shoulders **150**. The first and second underarm straps **134**, **136** are wrapped around the patient **122** and may be made to crisscross in the back of the patient **122**. The first ends **138**, **142** of the underarm straps **134**, **136** are connected at the lower end portion **152** of the chest brace **112** in front of the patient **122** and below the shoulders **150**. The second ends **140**, **144** of the underarm straps **134**, **136** are connected to the front collar **102** in the rear of the patient **122** and above the shoulders **150**. The underarm straps **134**, **136** may then be tightened snug against the patient's armpits **158**. The headband **104** may then be wrapped around the forehead **126** of the patient **122** and the lateral head supports **106**, **108** may be connected to the lateral collar projections **118**, **120** to provide a rigid connection between the headband **104** and front collar **102**.

[0046] Advantageously, even without the chin brace **110** connected to the CID **100**, the patient's neck **186** and head **184** are substantially immobilized at this point for several reasons. More specifically, the first and second shoulder pieces **114**, **116** provide a positive stop against the patient's shoulders **150** in the downward direction, while the first and second underarm straps **134**,

136 provide a positive stop against the patient's armpits **158** in the upwards direction. Therefore, once the rigid connection between the lateral head supports **106**, **108** and the lateral collar projections **118**, **120** is made, the distance **160** between the headband **104** and shoulder pieces **114**, **116** of the collar **102** is fixed.

[0047] Additionally, the chest brace **112** of the collar **102** is firmly anchored to the chest **124** of the patient **122** by the underarm straps **134**, **136** to prevent forward movement of the head **184** of the patient **122**. Also, the anchored chest brace **112** combined with the positive stop of the shoulder pieces **116**, **118** against the shoulders **150** prevent rearward movement of the head **184** of the patient **122**. Moreover, the crisscrossed underarm straps **134**, **136** substantially restrict rotational movement of the head **184** of the patient **122**.

[0048] For added support, the chin brace **110** may be removably attached to the front collar **102** to further immobilize the head **184** and neck **186** of the patient. The chin brace **110** may be pivoted at its first pivoting chin brace extensions **172** to comfortably abut against the chin of the patient **122** and then may be locked rigidly in place by the second adjusting chin brace extensions **174**. However, the chin brace **110** is the sole component of the CID **100** that is required to be removed from the CID **100**, in order to enable the patient **122** to be intubated. Advantageously, the CID, even with the chin brace **110** removed, will securely immobilize the patient **122** during the intubation process.

[0049] Referring to FIG. **4**, an example is depicted of a perspective view of the front collar **102**, which includes the chest brace **112** of the cervical immobilization device **100** of FIG. **1**, according to aspects described herein. The collar **102** is the main support component of the CID **100**.

[0050] The collar **102** provides several advantageous structural features for immobilizing and securing the head **184** and neck **186** of the patient **122** in a single integral structure. Those structural features include at least the following: [0051] the chest brace **112** of the collar **102**, for providing the leverage needed to prevent forward and rearward movement of the head **184** of the patient **122**; [0052] the first and second shoulder pieces **114**, **116** of the collar **102**, for providing a positive stop of the collar **102** against the shoulders **150** of the patient **122**; and [0053] the first and second lateral collar projections **118**, **120** of the collar **102**, for providing a rigid connection to the first and second lateral head supports **106**, **108** and ultimately a rigid connection to the headband **104** for securing the head **184** of the patient **122**.

[0054] The chest brace **112** of the collar **102** is designed to rest against the chest **124** of a patient **122** and provide the leverage needed to prevent forward and rearward movement of the head **184** of the patient **122**. The chest brace **112** utilizes the chest **124** as a positive stop against any forward movement of the head **184** of the patient **122**. Additionally, when the chest brace **112** is anchored to the chest **124** of the patient **122** by the underarm straps **134**, **136**, the chest brace **112** substantially prevents any rearward movement of the head **184** of the patient **122**.

[0055] The first and second shoulder pieces **114**, **116** of the collar **102** provide the positive stop against the shoulders **150** of the patient **122**. Additionally, the shoulder pieces **114**, **116** further restrict rearward movement of the patient's head **184**.

[0056] The first and second lateral collar projections **118**, **120** of the collar **102** are designed to adjustably mate with the lateral head supports **106**, **108** to provide the required rigid connection between the collar **102** and the headband **104**. Accordingly, once the collar **102** is anchored to the patient **122** by the underarm straps **134**, **136** and the headband **104** is rigidly connected to the collar **102** via the lateral head supports **106**, **108**, the head **184** and neck **186** of the patient **122** are restrained.

[0057] The lateral collar projections **118**, **120** are illustrated in FIG. **4** as having the adjustment tabs **162** disposed thereon, wherein they would mate with the adjustment thru-holes **164** on the lateral head supports **106**, **108**. Alternatively, however, the lateral collar projections **118**, **120** may also have the adjustment thru-holes **164** disposed thereon, wherein they would mate with the adjustment tabs **162** disposed on the lateral head supports **106**, **108**.

[0058] The front collar **102** is also illustrated in FIG. **4** as having the first tabs **176** and second tabs **180** disposed thereon. As such, the first tabs **176** would mate with the first thru-holes **178** disposed on the first pivoting chin brace extension **172** and the second tabs **180** would mate with the second thru-holes **182** disposed on the second adjusting chin brace extension **174** of the chin brace **110**. Alternatively, however, The front collar **102** may have the first and second thru-holes **178**, **182** disposed thereon. Accordingly, the first thru-holes **178** would mate with the first tabs **176** disposed on the first pivoting chin brace extension **172** and the second thru-holes **182** would mate with the second tabs **180** disposed on the second adjusting chin brace extension **174** of the chin brace **110**.
[0059] Referring to FIG. **5**, an example is depicted of a perspective view of the headband **104** and first and second lateral head supports **106**, **108** of the cervical immobilization device **100** of FIG. **1**, according to aspects described herein.
[0060] The headband **104**, as illustrated, wraps around a patient's forehead **126** and is adjustable via a hook and loop attachment system. However, any number of other attachment systems may also be used. For example, buckles, snaps, tabs or other fastening devices may be used to secure the headband **104** to the forehead **126** of the patient **122**.
[0061] The first and second lateral head supports **106**, **108** may be composed of a hard plastic material or any other appropriately rigid material that can adequately provide a rigid connection with the lateral collar projections **118**, **120** of the collar **102**.
[0062] As illustrated, the headband **104** feeds through the headband slots **132** on the head section **130** of the lateral head supports **106**, **108**. However, the lateral head supports **106** may be attached via any number of other appropriate attachment systems, such as via hook and loop, buckles, snaps, tabs or the like. Additionally, the lateral head supports **106**, **108** may be integrally and permanently attached to the headband **104**.
[0063] The lateral head supports **106**, **108** are illustrated as being adjustably attachable to the lateral collar projections **118**, **120** of the collar **102** via a plurality of adjustment tabs **162**. The adjustment tabs are configured to mate to a corresponding plurality of adjustment thru-holes **164**. More specifically, a plurality of headband adjustment tabs **162** may be positioned vertically on the first and second lateral head supports **106**, **108**. Additionally, a plurality of headband adjustment thru-holes **164** may be positioned vertically on the first and second lateral collar projections **118**, **120**. The headband adjustment thru-holes **164** are sized to receive the headband adjustment tabs **162**. During operation, one or more of the plurality of headband adjustment tabs **162** mate with (or press fit into) one or more of the plurality of headband adjustment thru-holes **164** to provide a rigid, but adjustable, connection between the first and second lateral head supports **106**, **108** and the first and second lateral collar projections **118**, **120** respectively. The adjustable connection enables the adjustment of the distance **160** between the headband **104** and the shoulder projections **114**, **116** of the front collar **102**. Alternatively, the plurality of headband adjustment tabs **162** may be positioned vertically on the first and second lateral collar projections **118**, **120** and the plurality of headband adjustment thru-holes **164** may be positioned vertically on the first and second head supports **106**, **108**.
[0064] The lateral head supports **106**, **108** are illustrated in FIG. **5** as having the adjustment tabs **162**, wherein they would mate with the adjustment thru-holes **164** on the lateral head supports **106**, **108**. Alternatively, however, the lateral collar projections **118**, **120** may also have the adjustment thru-holes **164** disposed thereon, wherein they would mate with the adjustment tabs **162** disposed on the lateral head supports **106**, **108**.
[0065] Though the lateral head supports **106**, **108** are illustrated as being adjustably attachable to the lateral collar projections **118**, **120** of the collar **102** via a plurality of adjustment tabs **162**, the lateral head supports **106**, **108** may be adjustably attached to the lateral collar projections **118**, **120** via any number of other attachment systems. For example, via hook and loop fasteners, buckles or the like.
[0066] Referring to FIG. **6**, an example is depicted of the chin brace **110** of the cervical

immobilization device **100** of FIG. **1**, according to aspects described herein. As discussed earlier herein, the chin brace **110** provides additional support to the head **184** of the patient **122** when attached to the CID **100**. The chin brace **110** may be made of a hard plastic material or other appropriate material that can provide a rigid brace against the chin of the patient **122**. The chin brace **110** may also include padding **128** to provide a more comfortable fit for the patient **122**.

[0067] Advantageously, the chin brace **110** is configured to be the sole component of the CID **100** that is required to be removed from the CID **100** to enable the patient **122** to be intubated while still wearing the CID **100**.

[0068] The chin brace **110** may be made removably attachable and/or pivotably attachable to the front collar **102** through any number of appropriate attachment systems. However, in the chin brace **110** illustrated herein, the attachment system of the chin brace **110** includes a first forked chin brace end portion **168** and a second forked chin brace end portion **170**. Each forked chin brace end portion **168**, **170** includes a first pivoting chin brace extension **172** pivotally and removably attached to the front collar **102**. Additionally, each forked chin brace end portion **168**, **170** includes a second adjusting chin brace extension **174** adjustably and removably attached to the front collar **102**.

[0069] More specifically, each first pivoting chin brace extension **172** includes one of a first tab **176** or a first thru-hole **178**, which is configured to mate with the other of the first tab **176** or the first thru-hole **178** positioned on the front collar **102**. In other words, if the first tab **176** is positioned on the first pivoting chin brace extension **172**, then the first thru-hole **178** will be positioned on the front collar **102**. Alternatively, if the first tab **176** is positioned on the front collar **102**, then the first thru-hole **178** will be positioned on the first pivoting chin brace extension **172**. In either case, the first tab **176** is configured to pivot within the first thru-hole **178** to enable the chin brace **110** to pivot relative to the front collar **102**.

[0070] Additionally, each second adjusting chin brace extension **174** includes one of a plurality second tabs **180** or a plurality of second thru-holes **182**, which are configured to mate with the other of the plurality of second tabs **180** or the plurality of second thru-holes **182** positioned on the front collar **102**. In other words, if the plurality of second tabs **180** are positioned on the second adjusting chin brace extension **174**, then the plurality of second thru-holes **182** will be positioned on the front collar **102**. Alternatively, if the plurality of second tabs **180** are positioned on the front collar **102**, then the plurality of second thru-holes **182** will be positioned on the second adjusting chin brace extension **174**. In either case, one or more of the plurality of second tabs **180** mate with one or more of the plurality of second thru-holes **182** to provide an adjustable connection between the second adjusting chin brace extensions **174** and the front collar **102** to enable adjustment of the pivot angle **166** between the chin brace **110** and the front collar **102**.

[0071] The chin brace **110** is illustrated in FIG. **6** as having the first thru-holes **178** disposed on the first pivoting chin brace extension **172** and the second thru-holes **182** disposed on the second adjusting chin brace extension **174**. As such, the first and second thru-holes **178**, **182** would mate with the first and second tabs **176**, **180** respectively, wherein the first and second tabs **176**, **180** would be mounted on the front collar **102**. Alternatively, however, the front collar **102** may have the first and second thru-holes **178**, **182** mounted thereon. Accordingly, the first and second tabs **176**, **180** would be mounted on the chin brace **110** and would mate with the first and second thru-holes **178**, **182** mounted on the front collar **102**.

[0072] In an alternative embodiment, the chin brace **110** may be a chin strap (not shown). The chin strap may be removably attached to the collar **102** by any number of appropriate fastening systems, such as hook and loop fasteners, snaps, tabs or the like.

[0073] Referring to FIG. 7A, an example is depicted of a partially exploded view of the cervical immobilization device **100** being worn by the patient **122** with the chin brace **110** separated from the CID **100**, according to aspects described herein.

[0074] The chin brace **110** may be easily removed from the patient **122** in order to provide access

to the mouth of the patient **122** and to enable the patient to be intubated. In the embodiments illustrated herein, the chin brace **110** may be removed by pulling on the first pivoting chin brace extensions **172** and the second adjusting chin brace extensions **174** to separate the first and second tabs **176**, **180** from the first and second thru-holes **178**, **182** respectively.
[0075] Referring to FIG. **7**B, an example is depicted of a perspective view of the CID **100** being worn by the patient **122** without a chin brace **110** while being intubated with an intubation tube **188**, according to aspects described herein.
[0076] The chin brace **110** is the sole component of the CID **100** that is required to be removed from the CID **100**, in order to enable the patient **122** to be intubated. Advantageously, the CID, even with the chin brace **110** removed, will securely immobilize the patient **122** during the intubation process, wherein the patient **122** may be intubated with an intubation tube **188**.
[0077] More specifically, even without the chin brace **110** attached to the CID **100**, the first and second shoulder pieces **114**, **116** provide a positive stop against the patient's shoulders **150** in the downward direction, while the first and second underarm straps **134**, **136** provide a positive stop against the patient's armpits **158** in the upwards direction. Therefore, once the rigid connection between the lateral head supports **106**, **108** and the lateral collar projections **118**, **120** is made, the distance **160** between the headband **104** and shoulder pieces **114**, **116** of the collar **102** is fixed.
[0078] Additionally, independent of the chest brace **110** being connected to the collar **102**, the chest brace **112** of the collar **102** is firmly anchored to the chest **124** of the patient **122** by the underarm straps **134**, **136** to prevent forward movement of the head **184** of the patient **122**. Also, the anchored chest brace **112** combined with the positive stop of the shoulder pieces **116**, **118** against the shoulders **150** prevent rearward movement of the head **184** of the patient **122**. Moreover, the crisscrossed underarm straps **134**, **136** substantially restrict rotational movement of the head **184** of the patient **122**.
[0079] It should be appreciated that all combinations of the foregoing concepts and additional concepts discussed in greater detail herein (provided such concepts are not mutually inconsistent) are contemplated as being part of the inventive subject matter disclosed herein. In particular, all combinations of claimed subject matter appearing at the end of this disclosure are contemplated as being part of the inventive subject matter disclosed herein.
[0080] Although the invention has been described by reference to specific examples, it should be understood that numerous changes may be made within the spirit and scope of the inventive concepts described. Accordingly, it is intended that the disclosure not be limited to the described examples, but that it have the full scope defined by the language of the following claims.

Claims

1. A cervical immobilization device (CID), comprising: a front collar comprising: a chest brace configured to extend downward from both shoulders of a patient and rest against a chest of the patient; an adjustable headband configured to strap around a forehead of the patient; first and second lateral head supports configured to attach and extend vertically downward from the headband, the first and second lateral head supports configured to form a rigid connection to the front collar; and first and second underarm straps each having first and second ends, wherein: the first ends are configured to attach to opposing sides of the chest brace below the patient's shoulders, and the second ends are configured to attach to opposing sides of the CID above the patient's shoulders when the CID is worn by the patient.
2. The cervical immobilization device of claim 1, wherein the second ends of the first and second underarm straps are configured to attach to the front collar.
3. The cervical immobilization device of claim 1, wherein the first and second underarm straps are configured to extend under and abut against a floor of the patient's armpits when the CID is worn by the patient.

4. The cervical immobilization device of claim 1, wherein: the first end of the first underarm strap is configured to attach to a left side of the chest brace and the second end of the first underarm strap is configured to attach to a right side of the CID; and the first end of the second underarm strap is configured to attach to a right side of the chest brace and the second end of the second underarm strap is configured to attach to a left side of the CID.
5. The cervical immobilization device of claim 1, wherein the front collar further comprises: first and second shoulder pieces integrally attached to opposing upper ends of the chest brace, the first and second shoulder pieces configured to project over and rest upon the shoulders of the patient to support the front collar thereon.
6. The cervical immobilization device of claim 1, wherein the front collar further comprises: first and second lateral collar projections integrally attached to opposing upper ends of the chest brace, the first and second lateral collar projections extending vertically upwards therefrom; wherein the first and second lateral collar projections are configured to adjustably attached to the first and second lateral head supports to adjust a distance between the headband and the front collar.
7. The cervical immobilization device of claim 6, comprising: a plurality of headband adjustment tabs positioned vertically on the first and second lateral head supports; and a plurality of headband adjustment thru-holes positioned vertically on the first and second lateral collar projections, wherein the headband adjustment thru-holes are sized to receive the headband adjustment tabs; wherein one or more of the plurality of headband adjustment tabs mate with one or more of the plurality of headband adjustment thru-holes to provide an adjustable connection between the first and second lateral head supports and the first and second lateral collar projections respectively to enable the adjustment of the distance between the headband and the front collar.
8. The cervical immobilization device of claim 6, comprising: a plurality of headband adjustment tabs positioned vertically on the first and second lateral collar projections; and a plurality of headband adjustment thru-holes positioned vertically on the first and second lateral head supports, wherein the headband adjustment thru-holes are sized to receive the headband adjustment tabs; wherein one or more of the plurality of headband adjustment tabs mate with one or more of the plurality of headband adjustment thru-holes to provide an adjustable connection between the first and second lateral head supports and the first and second lateral collar projections respectively to enable the adjustment of the distance between the headband and the front collar.
9. The cervical immobilization device of claim 1, comprising: a chin brace removably attached to opposing sides of the front collar; wherein the chin brace is configured to provide additional support to immobilize the patient, when the CID is being worn by the patient; and wherein the chin brace is configured to be the sole component of the CID required to be removed from the CID to enable the patient to be intubated while wearing the CID.
10. The cervical immobilization device of claim 9, wherein the chin brace is pivotally and adjustably attached to opposing sides of the front collar, to enable a pivot angle between the chin brace and the front collar to be adjusted.
11. The cervical immobilization device of claim 10, wherein the chin brace further comprises first and second forked chin brace end portions, each forked chin brace end portion comprising: a first pivoting chin brace extension pivotally and removably attached to the front collar; and a second adjusting chin brace extension adjustably and removably attached to the front collar.
12. The cervical immobilization device of claim 11, wherein: each first pivoting chin brace extension includes one of a first tab or a first thru-hole, which is configured to mate with the other of the first tab or the first thru-hole positioned on the front collar, wherein the first tab is configured to pivot within the first thru-hole to enable the chin brace to pivot relative to the front collar; and each second adjusting chin brace extension includes one of a plurality second tabs or a plurality of second thru-holes, which are configured to mate with the other of the plurality of second tabs or the plurality of second thru-holes positioned on the front collar, wherein one or more of the plurality of second tabs mate with one or more of the plurality of second thru-holes to provide an adjustable

connection between the second adjusting chin brace extensions and the front collar to enable adjustment of the pivot angle between the chin brace and the front collar.

13. A cervical immobilization device (CID), comprising: a front collar comprising: a chest brace configured to extend downward from both shoulders of a patient and rest against a chest of the patient; an adjustable headband configured to strap around a forehead of the patient; first and second lateral head supports configured to attach and extend vertically downward from the headband, the first and second lateral head supports configured to form a rigid connection to the front collar; and a chin brace removably attached to opposing sides of the front collar; wherein the chin brace is configured to provide additional support to immobilize the patient, when the CID is being worn by the patient; and wherein the chin brace is configured to be the sole component of the CID required to be removed from the CID to enable the patient to be intubated while wearing the CID.

14. The cervical immobilization device of claim 13, wherein the chin brace is pivotally and adjustably attached to opposing sides of the front collar, to enable a pivot angle between the chin brace and the front collar to be adjusted.

15. The cervical immobilization device of claim 14, wherein the chin brace further comprises first and second forked chin brace end portions, each forked chin brace end portion comprising: a first pivot chin brace extension pivotally and removably attached to the front collar; and a second adjusting chin brace extension adjustably and removably attached to the front collar.

16. The cervical immobilization device of claim 13, comprising: first and second underarm straps each having first and second ends, wherein: the first ends are configured to attach to opposing sides of the chest brace below the patient's shoulders, the second ends are configured to attach to opposing sides of the CID above the patient's shoulders, and the first and second underarm straps are configured to extend under and abut against a floor of the patient's armpits when the CID is worn by the patient.

17. The cervical immobilization device of claim 16, wherein: the first end of the first underarm strap is configured to attach to a left side of the chest brace and the second end of the first underarm strap is configured to attach to a right side of the CID; and the first end of the second underarm strap is configured to attach to a right side of the chest brace and the second end of the second underarm strap is configured to attach to a left side of the CID.

18. The cervical immobilization device of claim 13, wherein the front collar further comprises: first and second shoulder pieces integrally attached to opposing upper ends of the chest brace, the first and second shoulder pieces configured to project over and rest upon the shoulders of the patient to support the front collar thereon.

19. A cervical immobilization device (CID), comprising: a front collar comprising a single integral structure including the structural features of: a chest brace configured to extend downward from both shoulders of a patient and rest against a chest of the patient, first and second shoulder pieces integrally attached to opposing upper ends of the chest brace, the first and second shoulder pieces configured to project over and rest upon the shoulders of the patient to support the front collar thereon, and first and second lateral collar projections integrally attached to opposing upper ends of the chest brace, the first and second lateral collar projections extending vertically upwards therefrom; an adjustable headband configured to strap around a forehead of the patient; first and second lateral head supports configured to attach and extend vertically downward from the headband, the first and second lateral head supports configured to adjustably and rigidly attach to the first and second lateral collar projections to adjust a distance between the headband and the front collar; first and second underarm straps each having first and second ends, wherein: the first end of the first underarm strap is configured to attach to a left side of the chest brace below the patient's shoulder and the second end of the first underarm strap is configured to attach to a right side of the CID above the patient's shoulder; and the first end of the second underarm strap is configured to attach to a right side of the chest brace below the patient's shoulder and the second end of the second underarm strap is configured to attach to a left side of the CID above the patient's shoulder.

20. The cervical immobilization device of claim 19, comprising: a chin brace removably attached to opposing sides of the front collar; wherein the chin brace is configured to provide additional support to immobilize the patient, when the CID is being worn by the patient; and wherein the chin brace is configured to be the sole component of the CID required to be removed from the CID to enable the patient to be intubated while wearing the CID.